

## CPU vs GPU

Before GPUs became a thing, a CPU was the most vital component of a rendering setup - to such an extent that you didn't have to worry about any other stuff and just put all your money in procuring the best CPU possible, but alas, the times have changed. CPU rendering is no longer the only available option, and the focus has shifted largely in favour of the accelerators.

That said, the CPU is still vitally important. Besides being the brain of your system, in case of Blender, there are still times where you might need to use CPU for Cycles (Blender's very own render engine). For instance, while modelling, in theory and in practice you should only care about single-core performance so extra threads won't make much of a difference. GPU support for shadow-rendering is still pretty dicey or suppose you're doing large scenes that take up lots of RAM, you're going to be stuck with CPU, as GPUs are limited to their own VRAM. GPUs can't share VRAM either, so stacking GPUs won't increase their total memory. So, with Intel's arguably (Ryzen is a close competitor) best-in-the-market single core performance and overclocking chips you don't need to look any further.

This is not to say that your multi-threaded muscle power will go unused. A good GPU will go a long way in saving you tons of time on rendering. For rendering, unlike majority of the graphic design tasks, the graphics card's strength is the ultimate basis of determining how fast you're able to complete your work

GPU rendering is much faster—in some cases, being 10x as fast. You can go for either AMD or NVIDIA since Blender is optimised for both.

Between the two most developed kinds of GPU rendering available now (AMD-backed OpenCL and NVIDIA's proprietary CUDA), CUDA currently is ahead in performance if you look at the Benchmark results. And while OpenCL, to an extent is compatible with any graphics card, CUDA is only available with NVIDIA cards. To quote GPU Rendering Magazine's article on OpenCL and CUDA: "As an easy rule of thumb, if your app supports CUDA, grab an NVIDIA card, even if it also supports OpenCL," and "If the applications you use split their support between CUDA and OpenCL we recommend using a recent NVIDIA card."

Let's have a quick look at OctaneBench scores.

It is clear that a GTX 780 Ti will give you the best bang for your buck whereas if you are willing to shell out some more of those bills you can go for the RTX 2080 which scores almost double to that of its sibling 780 Ti.